# SMART RESUME SCREENING TOOL

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UNDER THE GUIDANCE OF  DR. P ANITHA

# PROBLEM STATEMENT:

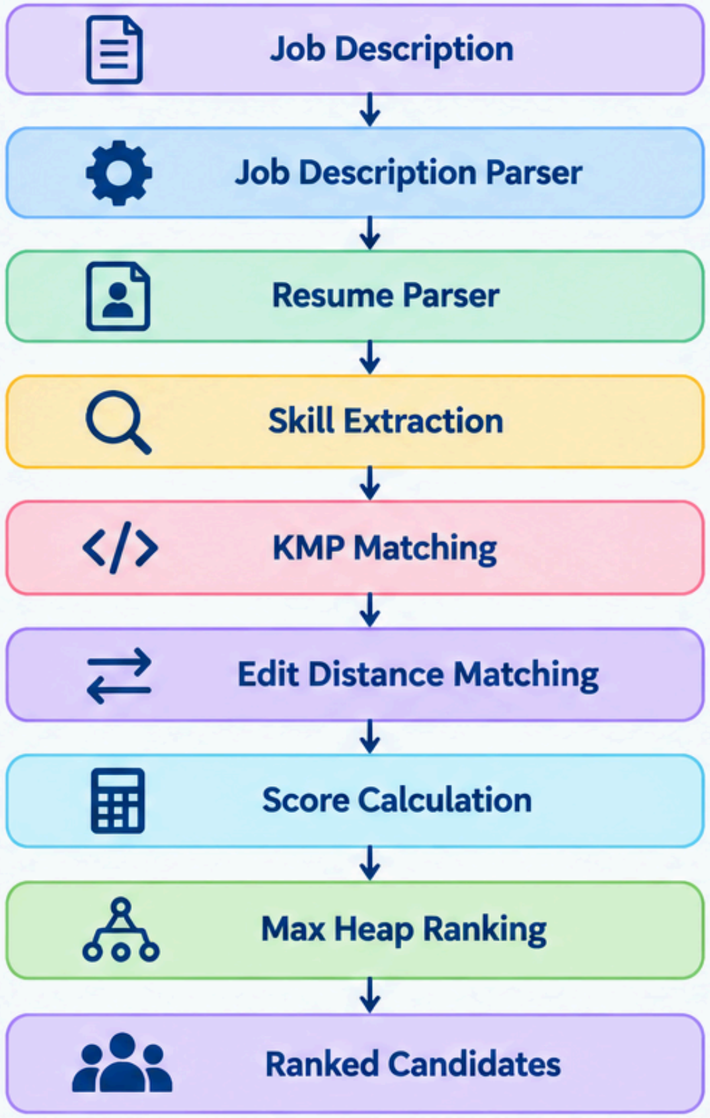
- Recruiters receive a large number of resumes for each job opening.
- Manual screening is time-consuming and inefficient.
- Important skills and qualifications may be overlooked.
- Comparing resumes with job requirements requires significant effort.
- An automated system is needed to analyze and rank resumes efficiently.

# OBJECTIVES:

- Extract skills, experience, education, and certifications from resumes.
- Match candidate profiles with job descriptions.
- Rank candidates based on relevance.
- Reduce recruiter workload and screening time.

# System Architecture & Workflow

## System Architecture




## Workflow

1. Read Job Description
2. Read Resume Files
3. Extract Skills & Experience
4. Match Skills using KMP
5. Handle Similar Skills using Edit Distance
6. Rank Candidates using Max Heap

# DSA CONCEPTS & TECHNOLOGY STACK

| Module | Algorithm |
| --- | --- |
| Exact Skill Matching | KMP Algorithm |
| Fuzzy Matching | Edit Distance (Dynamic Programming) |
| Candidate Ranking | Max Heap |
| Searching | String Search |
| Resume Parsing | File Processing |

# OUTPUT

## Sample Job Description

```
Software Developer

We are looking for a Software Developer with strong knowledge of Java, Python, SQL, Data Structures, and
Algorithms.

Preferred Skills:
Spring Boot, AWS, MySQL

Candidates should have at least 2 years of experience.

AWS certification is preferred.

Responsibilities:
- Develop scalable applications
- Design database systems
- Work with cloud platforms
```

# OUTPUT

## Sample Output

```
=====================================
 SMART RESUME SCREENING TOOL
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===== FINAL RANKING =====

Rank #1
Candidate: Rahul Sharma
Skills: Java Python SQL Spring Boot AWS Data Structures Algorithms JavaScript HTML CSS MySQL C
Experience: 2 years
Certifications: AWS
Score: 115

Rank #2
Candidate: Sai Kumar
Skills: Python SQL AWS Machine Learning Data Structures Algorithms MongoDB C
Experience: 3 years
Certifications: AWS
Score: 95

Rank #3
Candidate: Ananya Reddy
Skills: Java Python SQL JavaScript HTML CSS React NodeJS MySQL C
Experience: 1 years
Certifications: Oracle
Score: 55
```

# RESULTS & FUTURE SCOPE

## Results

- Automatically analyzes resumes.
- Matches candidate skills with job requirements.
- Ranks candidates based on suitability.
- Reduces manual screening effort.

## Future Scope

- PDF Resume Support (PDFBox)
- Resume Upload GUI
- MySQL Database Integration
- Web Application Deployment

# CONCLUSION

The Smart Resume Screening Tool successfully demonstrates the application of DSA concepts such as KMP, Dynamic Programming (Edit Distance), and Max Heap to automate resume analysis and candidate ranking.

# THANK YOU